Republic of the Philippines
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LAB REPORT

(CIRCULAR SINGLY LINKED LIST)

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CIRCULAR SINGLY LINKED LIST

MACHINE PROBLEM DESCRIPTION:

This program focuses on the operations of a Circular Singly Linked List, a data structure where each node contains data and a reference to the next node. Unlike a regular Singly Linked List, the last node in a Circular Singly Linked List points back to the head node, forming a circular structure. Operations include adding nodes to the end (append), beginning (prepend), deleting nodes, and traversing the list to perform actions like printing data. This program helps us understand how Circular Singly Linked Lists work and their applications in solving real-world problems, particularly where circular data structures are advantageous.

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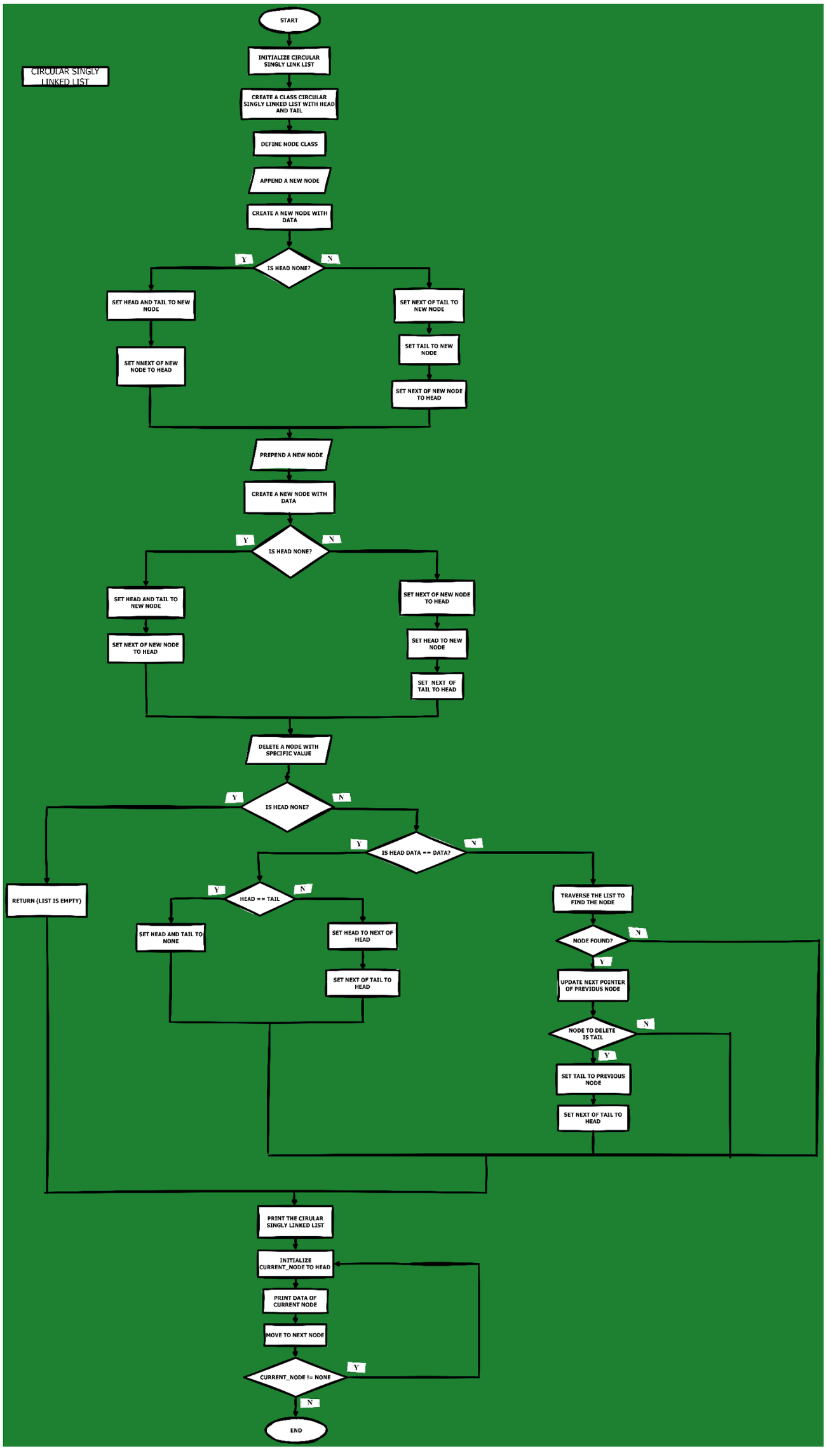
FLOWCHART (CIRCULAR SINGLY LINKED LIST):

This flowchart represents a scenario where the program processes a Circular Singly Linked List. The flowchart details the step-by-step process.

``


CIRCULAR SINGLY LINKED LIST
START
INITIALIZE CIRCULAR SINGLY LINK LIST
CREATE A CLASS CIRCULAR SINGLY LINKED LIST WITH HEAD AND TAIL
DEFINE NODE CLASS
APPEND A NEW NODE
CREATE A NEW NODE WITH DATA
IS HEAD NONE?
Y
N
SET HEAD AND TAIL TO NEW NODE
SET NNEXT OF NEW NODE TO HEAD
SET NEXT OF TAIL TO NEW NODE
SET TAIL TO NEW NODE
SET NEXT OF NEW NODE TO HEAD
PREPEND A NEW NODE
CREATE A NEW NODE WITH DATA
IS HEAD NONE?
Y
N
SET HEAD AND TAIL TO NEW NODE
SET NEXT OF NEW NODE TO HEAD
SET NEXT OF NEW NODE TO HEAD
SET HEAD TO NEW NODE
SET NEXT OF TAIL TO HEAD
DELETE A NODE WITH SPECIFIC VALUE
IS HEAD NONE?
Y
N
RETURN (LIST IS EMPTY)
IS HEAD DATA == DATA?
Y
N
HEAD == TAIL
Y
N
SET HEAD AND TAIL TO NONE
SET HEAD TO NEXT OF HEAD
SET NEXT OF TAIL TO HEAD
TRAVERSE THE LIST TO FIND THE NODE
NODE FOUND?
N
Y
UPDATE NEXT POINTER OF PREVIOUS NODE
NODE TO DELETE IS TAIL
N
Y
SET TAIL TO PREVIOUS NODE
SET NEXT OF TAIL TO HEAD
PRINT THE CIRULAR SINGLY LINKED LIST
INITIALIZE CURRENT_NODE TO HEAD
PRINT DATA OF CURRENT NODE
MOVE TO NEXT NODE
CURRENT_NODE != NONE
Y
N
END

SOURCE CODE (CIRCULAR SINGLY LINKED LIST):

The code below demonstrates the implementation of various operations on a Circular Singly Linked List of integers.

``

```python
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class CircularSinglyLinkedList:
    def __init__(self):
        self.head = None
        self.tail = None

    def append(self, data):
        new_node = Node(data)
        if not self.head:
            self.head = new_node
            self.tail = new_node
            new_node.next = self.head
        else:
            self.tail.next = new_node
            self.tail = new_node
            self.tail.next = self.head

    def prepend(self, data):
        new_node = Node(data)
        if not self.head:
            self.head = new_node
            self.tail = new_node
            new_node.next = self.head
        else:
            new_node.next = self.head
            self.head = new_node
            self.tail.next = self.head

    def delete(self, data):
        if not self.head:
            return "List is empty"

        if self.head.data == data:
            if self.head == self.tail:
                self.head = None
                self.tail = None
            else:
                self.head = self.head.next
                self.tail.next = self.head
            return

        current = self.head
        while current.next != self.head:
            if current.next.data == data:
                current.next = current.next.next
                if current.next == self.head:
                    self.tail = current
                return
            current = current.next

    def print_list(self):
        if not self.head:
            return "List is empty"

        current = self.head
        while True:
            print(current.data, end=" -> ")
            current = current.next
            if current == self.head:
                break
        print()

# Example usage
csll = CircularSinglyLinkedList()
csll.append(1)
csll.append(2)
csll.append(3)
csll.prepend(0)
csll.print_list()  # 0 -> 1 -> 2 -> 3 ->
csll.delete(2)
csll.print_list()  #0 -> 1 -> 3 ->
```

OUTPUT: (CIRCULAR SINGLY LINKED LIST):

The output displays the results of various operations on the Circular Singly Linked List

```
[Running] python -u "c:\Users\Gian Carlo Trilles\OneDrive\Desktop\pythonfiles\CircularSinglyLinkList.py"
0 -> 1 -> 2 -> 3 ->
0 -> 1 -> 3 ->

[Done] exited with code=0 in 0.091 seconds
```

``

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